EXISTENCE OF A SECTION OF O(1) GIVING AN INJECTIVE MULTIPLICATION F(-1) -> F WITH COKERNEL PUSHED FORWARD FROM THE ASSOCIATED HYPERPLANE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Associated primes of a finite module over a Noetherian ring and the zero-divisor criterion. Stacks Project, Algebra, Lemmas algebra-lemma-finite-ass (tag 00LC) and algebra-lemma-ass-zero-divisors (tag 00LD). Finiteness of Ass(M) and the fact that an element outside all associated primes is a non-zero-divisor drive the injectivity argument.
- Multiplication by a section on a quasi-coherent sheaf. Stacks Project, Constructions, Equation constructions-equation-multiply-on-sheaf (tag 03GJ). Defines exactly the map F(-1) -> F used in the statement.
- The zero scheme of a section of an invertible sheaf and i_*-equivalence on the hyperplane. Stacks Project, Divisors, Definition divisors-definition-zero-scheme-s (tag 02OQ) and Cohomology of Schemes, Lemma coherent-lemma-i-star-equivalence (tag 087T). Identify the ideal sheaf of H and descend the cokernel to the hyperplane.

Source reference. Original reference 03GJ: sources/stacks-chow/tags.txt, source lines 4530–4530. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00LD: sources/stacks-chow/tags.txt, source lines 783–783. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00LC: sources/stacks-chow/tags.txt, source lines 782–782. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OQ: sources/stacks-chow/tags.txt, source lines 3525–3525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 087T: sources/stacks-chow/tags.txt, source lines 10438–10438. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Lemma 0.1. *Let k be an infinite field. Let $n \geq 1$. Let $\mathcal{F}$ be a coherent module on $\mathbf{P}^n_k$. Then there exist a nonzero section $s \in \Gamma(\mathbf{P}^n_k, \mathcal{O}(1))$ and a short exact sequence*

$$0 \to \mathcal{F}(-1) \to \mathcal{F} \to i_*\mathcal{G} \to 0$$

where $i : H \to \mathbf{P}^n_k$ is the hyperplane H associated to s and $\mathcal{G} = i^*\mathcal{F}$.

Proof. The map $\mathcal{F}(-1) \to \mathcal{F}$ comes from Constructions, Equation (03GJ) with $n = 1$, $m = -1$ and the section s of $\mathcal{O}(1)$. Let's work out what this map looks like if we restrict it to $D_+(T_0)$. Write $D_+(T_0) = \mathrm{Spec}(k[x_1, \ldots, x_n])$ with $x_i = T_i/T_0$. Identify $\mathcal{O}(1)|_{D_+(T_0)}$ with $\mathcal{O}$ using the section T_0. Hence if $s = \sum a_i T_i$ then $s|_{D_+(T_0)} = a_0 + \sum a_i x_i$ with the identification chosen above. Furthermore, suppose $\mathcal{F}|_{D_+(T_0)}$ corresponds to the finite $k[x_1, \ldots, x_n]$-module M. Via the identification $\mathcal{F}(-1) = \mathcal{F} \otimes \mathcal{O}(-1)$ and our chosen trivialization of $\mathcal{O}(1)$ we see that $\mathcal{F}(-1)$ corresponds to M as well. Thus restricting $\mathcal{F}(-1) \to \mathcal{F}$ to $D_+(T_0)$ gives the map

$$M \xrightarrow{a_0 + \sum a_i x_i} M$$

To see that the arrow is injective, it suffices to pick $a_0 + \sum a_i x_i$ outside any of the associated primes of M, see Algebra, Lemma 00LD. By Algebra, Lemma 00LC the set $\mathrm{Ass}(M)$ of associated primes of M is finite. Note that for $\mathfrak{p} \in \mathrm{Ass}(M)$ the intersection $\mathfrak{p} \cap \{a_0 + \sum a_i x_i\}$ is a proper k-subvector space. We conclude that there is a finite family of proper sub vector spaces $V_1, \ldots, V_m \subset \Gamma(\mathbf{P}^n_k, \mathcal{O}(1))$ such that if we take s outside of $\bigcup V_i$, then multiplication by s is injective over $D_+(T_0)$. Similarly for the restriction to $D_+(T_j)$ for $j = 1, \ldots, n$. Since k is infinite, a finite union of proper sub vector spaces is never equal to the whole space, hence we may choose s such that the map is injective. The cokernel of $\mathcal{F}(-1) \to \mathcal{F}$ is annihilated by $\mathrm{Im}(s : \mathcal{O}(-1) \to \mathcal{O})$ which is the ideal sheaf of H by Divisors, Definition 02OQ. Hence we obtain $\mathcal{G}$ on H using Cohomology of Schemes, Lemma 087T. □